

Where We Sit: The Credit-Mediated Transition into Family-Sized Ownership

The Mechanism: Credit $\rightarrow$ Ownership $\rightarrow$ Space $\rightarrow$ Child

Hacamo (2021, *RFS*) identifies a causal chain in the data:

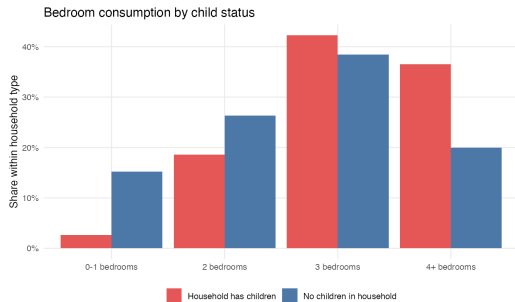
$$\text{credit access} \rightarrow \underbrace{\text{cross DP threshold}}_{b \geq (1-\phi) p_i H_k} \rightarrow \text{ownership of a larger unit} \rightarrow \text{child}$$

The killer test (his Table 8). He splits the joint outcome $\mathbf{1}\{\text{buy} \cap \text{newborn}\}$ by the size of the purchased home:

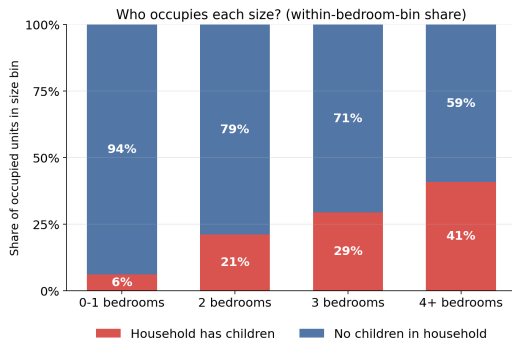
$y = \mathbf{1}\{\text{Mortgage} \& \text{newborn} \& [\text{size}]\}$	Large (4+ rooms)	Small (<4 rooms)
APL $\times$ Post $\times$ Fraction OCC	0.05 to 0.06**	0.01 (n.s.)

The credit-deregulation effect on babies works *only* through purchases of bigger homes. The chain is not generic; it is the family-space chain. Supplemental tests reject income growth and housing-wealth growth as alternatives.

Families Need Big Units — Two Views



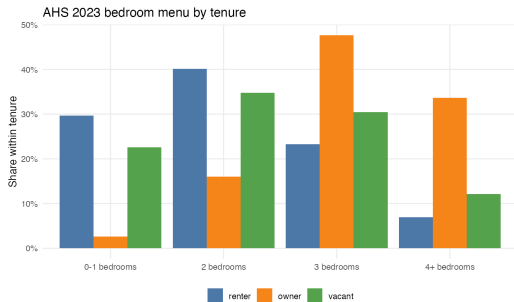
Within each household type: of households with kids, 79% live in 3+ bedroom units; of childless, 59%.



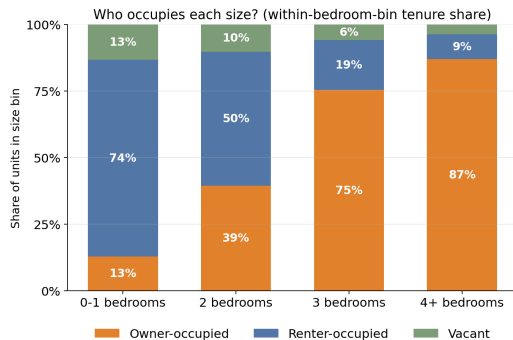
Within each size bin: share occupied by parents rises monotonically from 6% (0–1 bed) to 41% (4+ bed).

Two sides of the same fact, same data: families need bigger units, and the bigger-unit stock is dominated by families. The gradient on the right (6 → 21 → 29 → 41% parents) is the cleanest expression — it says directly who occupies the family-sized stock. Source: AHS 2023.

Big Units Are Owned — Renting Is Not a Substitute



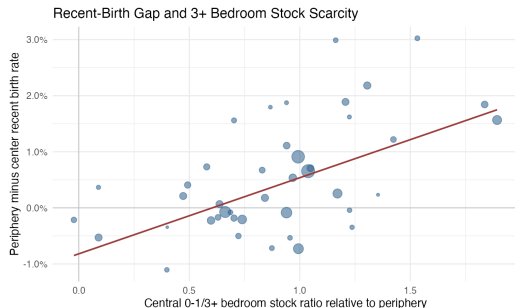
Within tenure: of owners, 81% live in 3+ bedroom units; of renters, 30%.



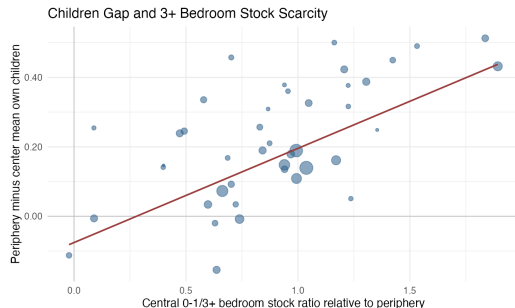
Within size bin: owner share of occupied units rises 13 → 39 → 75 → 87% as bedrooms grow. Renter share falls 74 → 50 → 19 → 9%.

The within-size view (right) is the cleanest version: of all 4+ bedroom homes, 87% are owner-occupied. The family-sized stock is essentially the owner-occupied stock. Reaching the family-sized margin requires owning, not paying more rent — the structural asymmetry our model captures via $\max H_k > \bar{h}_R$. Source: AHS 2023.

Across Metros: Where Family-Sized Stock Diverges, So Does Fertility



Periphery–center *recent-birth* gap rises with central 3+ bedroom stock scarcity.

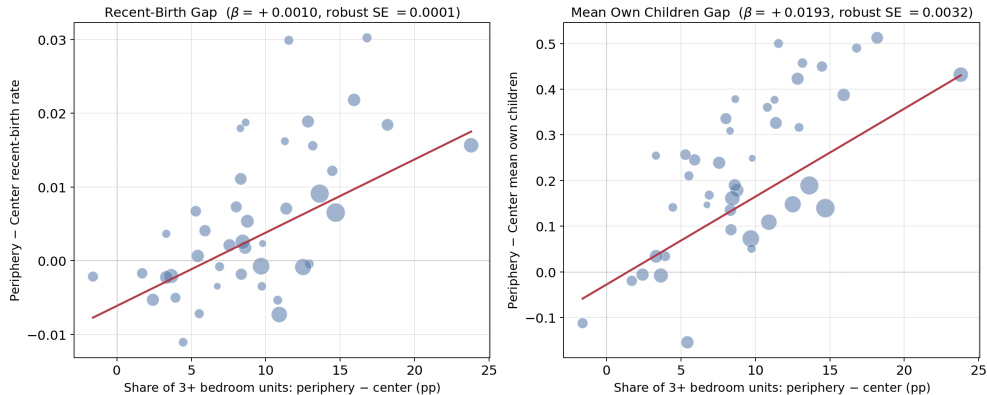


Periphery–center *mean own children* gap rises with the same scarcity measure.

ACS 2012–2023, 42 large MMS metros. x-axis: log of the central $\frac{0-1 \text{ bed}}{3+ \text{ bed}}$ stock ratio relative to periphery (higher = central family-sized stock is scarcer, Couillard-style measure). y-axes: periphery – center on the row's outcome.

The same axis predicts *parent centrality gap* with slope 0.147 (SE 0.026, $p < 10^{-6}$). Robust to expanding the sample to 80 metros (top-100-c25).

Same Pattern, Cleanest Availability Measure: Share of Family-Sized Stock



ACS 2012–2023, 42 large MMS metros. *x*-axis: *share of 3+ bedroom units in periphery – share in center*, in percentage points (mean 9.2, range –1.6 to 23.8). *y*-axes: periphery – center on the row's outcome.

Each +1 pp in the family-stock share gap is associated with ~ 0.1 pp on the recent-birth gap and 0.02 on the mean-children gap (both robust SE small enough to clear conventional thresholds). The mean-rooms-gap and bedroom-scarcity log-ratio versions give the same picture in different units.

Our PSID: Families Space Out at Birth

The destination side of Hacamo's chain shows up cleanly in PSID. Event study around first birth, rooms as outcome:

	First birth	Second birth
Pre-event mean (rooms)	6.01	6.14
α_{+3} (rooms at +3 yrs)	+ 0.66 (SE 0.15)	+ 0.70 (SE 0.18)
α_{+5} (rooms at +5 yrs)	+0.84 (0.15)	+0.72 (0.18)
<i>N</i> obs / IDs	248,566 / 23,041	155,358 / 15,568

Both first and second births add roughly 0.7 rooms by year +3. Babies arrive with bigger homes. Combined with Hacamo's Table 8 and the AHS asymmetry, the chain is end-to-end: families need big units, big units are owned, getting them requires the credit-mediated transition.

The Wealth Gate: Renters Who Can't Clear the DP Stay Childless

Among PSID renters at ages 25–30, pre-first-birth ($N = 2,961$, followed to age 45). Define DP shortfall at the FHA 3.5% floor:

$$\text{shortfall}_i = \mathbf{1}\{\text{liquid wealth}_i < 0.035 \cdot P_t^{\text{target}}\}.$$

Shortfall rate		$y = \text{childless by 45}$	$\hat{\beta}$	SE	p
Bottom income tercile	90%	Shortfall	0.088	0.029	0.003
Middle income tercile	83%	+ log income	0.088	0.029	0.003
Top income tercile	84%	+ log income + log NW	0.088	0.029	0.003

90% of low-income young renters lack the cash for a 3.5% FHA down payment on a typical first home. DP-short renters are 8.8 pp more likely to remain childless by 45, controlling for log income and log wealth. The DP gate predicts fertility, not the income level alone.

Even most top-income young renters fall short.

PSID 1984–2019. P_t^{target} = median first-time-buyer home value in same year-bin.

Existing Models Miss the Credit-Mediated Entry; Our Model Has It

	Couillard (2025)			Cumming & Dettling (2024)	Our model		
Tenure regime	owner	/	renter as states	owners only (existing mortgage holders)	owner	/	renter as states
Family-sized unit margin	bedrooms	1, 2, 3+		not modeled	discrete owner sizes H_k with $\max H_k > \bar{h}_R$		
Credit / DP constraint	absent:	no ϕ , no house prices, no liquid wealth state		not modeled; uses ARM/FRM payment shocks instead	$b \geq (1 - \phi) p_i H_k$ with $\phi = 0.80$ (DUE standard)		
What an APL/OCC credit shock would do	no model object responds			changes monthly payment, not the rent-to-own transition	relaxes the threshold; marginal renters cross into a larger H_k		

Couillard can describe the destination but not the credit trigger that lets a marginal renter reach it. Cumming & Dettling identify a real fertility response to mortgage credit but on the post-ownership cash-flow margin, not the entry margin. We are the only structural framework where Hacamo's specific causal chain has a model object behind every arrow.

What This Buys

Three structural ingredients deliver the Hacamo chain:

1. Discrete tenure choice — rent or own as distinct regimes.
2. Collateral constraint $b \geq (1 - \phi) p_i H_k$, $\phi = 0.80$ following DUE.
3. Tenure-asymmetric housing services: $\max H_k > \bar{h}_R$, so the family-sized units AHS shows in the data are reachable only through ownership.

What this lets us do that Couillard's framework cannot:

- Structural quantification of Hacamo's reduced-form chain. The fertility response to a credit shock has an explicit decomposition into DP-threshold crossings and post-crossing housing-size choices.
- A credit-side counterfactual portfolio: DP relief ($\phi \uparrow$), FHA-style threshold expansion, parent-targeted credit subsidies. These pull a lever Couillard's supply-only model does not have.
- A natural connection to the spatial dimension: p_i varies across center and periphery, so the credit gate binds differently across locations and ties cleanly into the parent-vs-non-parent sorting fact in MMS.